

B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Third & Fourth Semester

21CSC202J - OPERATING SYSTEMS

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

- i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours**Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer all Questions

Marks BL CO

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|----|--|---|---|---|---|
| 1. | Which among the following is the fastest among all the categories in a memory hierarchy
(A) Main memory
(C) CPU registers | (B) L1 Cache memory
(D) Secondary Memory | 1 | 2 | 1 |
| 2. | Device controller informs CPU that it has finished its operation by causing _____
(A) Interrupt
(C) Information | (B) Message
(D) Command | 1 | 1 | 1 |
| 3. | Which among the following service is provided by Operating System to serve the user?
(A) Resource Allocation
(C) Logging | (B) Protection and Security
(D) File System Manipulation | 1 | 2 | 1 |
| 4. | In a multiprogramming system, what is the primary goal?
(A) To ensure a single program runs faster
(C) To minimize user input | (B) To maximize CPU utilization
(D) To provide an integrated development environment | 1 | 2 | 1 |
| 5. | The processes that are residing in main memory and are ready and waiting to execute are kept on a list called _____
(A) job queue
(C) device queue | (B) ready queue
(D) process queue | 1 | 1 | 2 |
| 6. | Which of the following two operations are provided by the IPC facility?
(A) write & delete message
(C) send & receive message | (B) delete & receive message
(D) send & delete message | 1 | 1 | 2 |
| 7. | Which of the following schedulers is the fastest?
(A) Medium Term Scheduler
(C) Short Term Scheduler | (B) Job Scheduler
(D) Long term Scheduler | 1 | 1 | 2 |
| 8. | Each process identified and managed via a _____
(A) Process Control Block
(C) Device Queue | (B) Process Identifier
(D) Scheduling ID | 1 | 2 | 2 |

9. What is the main difference between soft real-time systems and hard real-time systems? 1 2 3
- (A) Soft real-time systems can miss deadlines occasionally without severe consequences, whereas hard real-time systems must meet all deadlines.
- (B) Soft real-time systems guarantee that processes will be given preference over noncritical processes, whereas hard real-time systems guarantee deadline adherence.
- (C) Hard real-time systems provide no guarantee as to when a critical real-time process will be scheduled, while soft real-time systems guarantee deadline adherence.
- (D) Soft real-time systems and hard real-time systems have no significant differences in scheduling.
10. What does the term "throughput" mean in scheduling terms? 1 1 3
- (A) The number of processes completed per unit time
- (B) The amount of time the CPU is idle
- (C) The total time a process spends in the system
- (D) The total time taken from process submission to completion
11. Regarding the Multilevel Queue Scheduling algorithm, which of the following is true? 1 2 3
- (A) Processes are permanently assigned to a queue
- (B) Each queue has its own scheduling algorithm
- (C) Processes can move between queues
- (D) Both a and b
12. In the deadlock system model, what is meant by "mutual exclusion"? 1 1 3
- (A) Processes hold resources while waiting for additional resources.
- (B) Resources cannot be forcibly taken from processes holding them.
- (C) Only one process can use a resource at a time.
- (D) A circular chain of processes exists, each holding a resource the next process needs.
13. Which of the following page replacement algorithms suffers from Belady's anomaly? 1 1 4
- (A) FIFO
- (B) LRU
- (C) Optimal Page Replacement
- (D) Both LRU and FIFO
14. Which of the following statements is false? 1 2 4
- (A) Virtual memory implements the translation of a program's address space into physical memory address space
- (B) Virtual memory reduces the degree of multiprogramming
- (C) Virtual memory allows each program to exceed the size of the primary memory
- (D) Virtual memory leads to/increases the context switching overhead
15. Which of the following is/are the strengths of virtual memory segmentation techniques used in memory management. 1 2 4
- i) No internal fragmentation
- ii) A higher degree of multiprogramming
- iii) More efficient to use of main memory
- iv) Large virtual address space
- v) Protection and sharing support
- (A) i, ii, iii and iv only
- (B) i, ii, iii and v only
- (C) i, ii, iv and v only
- (D) ii, iii, iv and v only

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|--|---|---|---|
| 16. Consider a machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4KB, what is the approximate size of the page table? | 1 | 5 | 4 |
| (A) 2 MB | (B) 8 MB | | |
| (C) 16 MB | (D) 24 MB | | |
| 17. For system protection, a process should access _____ | 1 | 1 | 5 |
| (A) all the resources | (B) only those resources for which it has authorization | | |
| (C) few resources but authorization is not required | (D) all of the mentioned | | |
| 18. Which one of the following is capability based protection system? | 1 | 1 | 5 |
| (A) hydra | (B) cambridge CAP system | | |
| (C) both hydra and cambridge CAP system | (D) none of the mentioned | | |
| 19. Access matrix model for user authentication contains _____ | 1 | 1 | 5 |
| (A) a list of objects | (B) a list of domains | | |
| (C) a function which returns an object's type | (D) All of the mentioned | | |
| 20. The pattern that can be used to identify a virus is known as _____ | 1 | 2 | 5 |
| (A) stealth | (B) virus signature | | |
| (C) armoured | (D) multipartite | | |

PART - B (5 × 8 = 40 Marks)

Answer all Questions

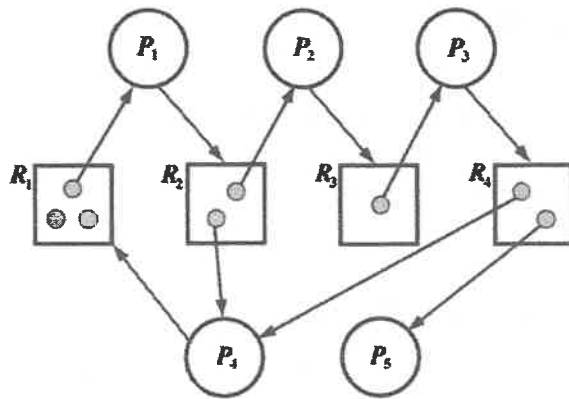
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|---|---|---|---|
| 21. (a) Consider that you are a software developer working on an operating system project. You need to interact with the kernel to perform various tasks like reading from files, writing to files, creating processes, and managing memory. These interactions are done using system calls. You need to explain to a new team member what system calls are and describe the different types available in detail. | 8 | 3 | 1 |
| (OR) | | | |
| (b) How could a system be designed to allow a choice of operating systems from which to boot? What would the bootstrap program need to do? | | | |
| 22. (a) Elaborate the actions taken by the kernel to context-switch between processes. Sketch the steps in CPU Switching From Process to Process. | 8 | 2 | 2 |
| (OR) | | | |
| (b) Paraphrase the role of the kernel in an operating system and how it interacts with hardware and software. | | | |
| 23. (a) Explain the concept of CPU scheduling and describe the criteria used to evaluate different CPU scheduling algorithms. | 8 | 2 | 3 |
| (OR) | | | |
| (b) Explain the necessary conditions for deadlock to occur and discuss the methods for deadlock prevention. | | | |
| 24. (a) What are the different accessing methods of a file? | 8 | 4 | 4 |
| (OR) | | | |
| (b) Explain the role and significance of the LDT and GDT in memory segmentation and how they help in managing memory efficiently. | | | |
| 25. (a) Define and explain the different classifications of computer security. | 8 | 2 | 5 |
| (OR) | | | |
| (b) Explain the concept of protection mechanism in operating systems and illustrate the use of protection domain and access control list with suitable examples. | | | |

PART - C (1 × 15 = 15 Marks)

26. Consider the following resource allocation graph.

15 3 1



(a) Convert it to the matrix representation (i.e., Allocation, Request and Available). [5 Marks]

(b) Do a step-by-step execution of the deadlock detection algorithm. For each step, add and remove the directed edges, and redraw the resource allocation graph. [7 Marks]

(c) Is there a deadlock? If there is a deadlock, which processes are involved? [3 Marks]

27. Consider a disk queue with requests for I/O to blocks on cylinders

15 4 4

93, 183, 37, 122, 14, 124, 65, 67

If the disk head is start at 53, then find out the total head movement with respect to

(i) FCFS [3 Marks],

(ii) SSTF [3 Marks],

(iii) SCAN [3 Marks],

(iv) C-SCAN [3 Marks] and

(v) LOOK scheduling [3 Marks]

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